Harsh Kumar Swami

9319336237, Email: harsh21149@iiitd.ac.in
Address: Janakpuri, New Delhi-59
Github , Linkedin

Education

Indraprastha Institute of Information Technology, Delhi
B.Tech (ECE)
2021-2025 (Present)
DAV Janakpuri, Delhi
(CBSE) 2019-2020
DAV Narela, Delhi
(CBSE) 2017-2018

Skills

Skill Set	Leadership and team management, creative thinking, problem-solving, project management, and analytical thinking.
Programming Language	Java, Python, C++, C, JavaScript, SQL and Verilog.
Tools and Technologies	EDA Tool, DSO, Git, MYSQL, MATLAB, HTML, Pandas, Numpy, OpenSTA, Cadence, TinkerCad, Smith Cart.
Technical Electives	Database Management System, Computer Network, Analog and Digital, Data Structures and Algorithms.
Core Elective	Digital Circuits, Basic Electronics, Embedded Logic Design, Signal & System, Circuit Theory and Devices, Digital Signal Processing, PCB Fabrication, Applied Fields and Waves, Integrated Circuits, Analog CMOS, Verilog Design, Radar System.

Work Experience

Cad Tool Development (B.Tech Project) Research (Sep 23–May 24)
Guide: Sayak Bhattacharya

- Semiconductor devices to develop the CAD tool development in solid states.
- Devices such as solar cells and quantum wells for 2D-3D shapes of cells.
- Different type of meshing technique. And try to apply that on the 2D and 3D shapes of a figure to see the small changes and with that different method of refinement.
- we have to make a small grid, we have to analyse the change in parameter with change in different grid.
- Analyze the change of any quantity like in semiconductors we have resistance and voltage. We can see the change in this quantity by grids.

Projects

Scan cells (Sep,24-Nov,24)
Project link

- A Verilog **netlist** for circuit design. Write an SDC file with clock period and clock uncertainty.
- Run STA on the written netlist and find slack for the worst time for the late and early path between F1 & F2, where F1, F2, and F3 are scan cells.
- Writing an SDC file for functional and shift mode and adding clock latency and changing uncertainty observes the change in the slack of early and late paths.

Differential Amplifier with current mirror (Oct,24-Dec 24)
Project link

- Designing and optimizing the differential amplifier with current mirrors in the circuits.
- To emphasize achieving high accuracy and low distortion by identifying and finding the threshold voltage and sizing of all Mosfets, optimizing parameters for good performance, and improving the CMRR.
- Creation, component sizing, and analysis of AC, frequency, and DC responses.

Mini SPY Device (Pocket Listening Device) (Mar,23-May,23)
Pocket listening

- A small signal input passes through a mic with a common emitter configuration BJT amplifier with the attenuator and the CE configuration BJT amplifier and with the attenuator.
- The CE configuration BJT amplifier and the attenuator and the CE configuration BJT amplifier and the Audio Jack in the plugin to pass the signal and to get the output through earphones to hear the signal and noise.

Positions of Responsibility

- Cyfuse club- Project and Photography head Jun,23
- Finnexia Videography and Photography Head Mar,23

Awards and Achievements

- Completed a Self-growth course in Swimming
- Completed community work-Summer Camp by a teacher of physics and literature.
- Run 3-4 Marathons (21 km, 10 km, 7 km)

Interests and Hobbies

- Badminton and Cricket
- Puzzle Solving and Marathon.

Declaration: The above information is correct to the best of my knowledge.
Harsh Kumar Swami
Date: April 16, 2025.